MIL-I-27623C
AMENDMENT 3
6 JUNE 1986
SUPERSEDING
AMENDMENT 2
26 JUNE 1980

MILITARY SPECIFICATION

INDICATOR, ATTITUDE, REMOTE STANDBY, 2 INCH

This amendment forms a part of Military Specification MIL-I-27623C, dated 10 January 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2 Add the following:

"MS27150-6 White intergral lighting with bank angle scale at bottom and with improved antireflection characteristics. Similar to -3 except all lamp position to be equipped with white lamps."

2.1 Under SPECIFICATION Federal: Delete "QQ-P-416" and add "PPP-B-601,— Boxes, Wood, Cleated-Plywood".

Add boxed paragraph to bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

2.1 Under SPECIFICATION Military add the following: " MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited".

PAGE 3

Add "3.4.2.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item."

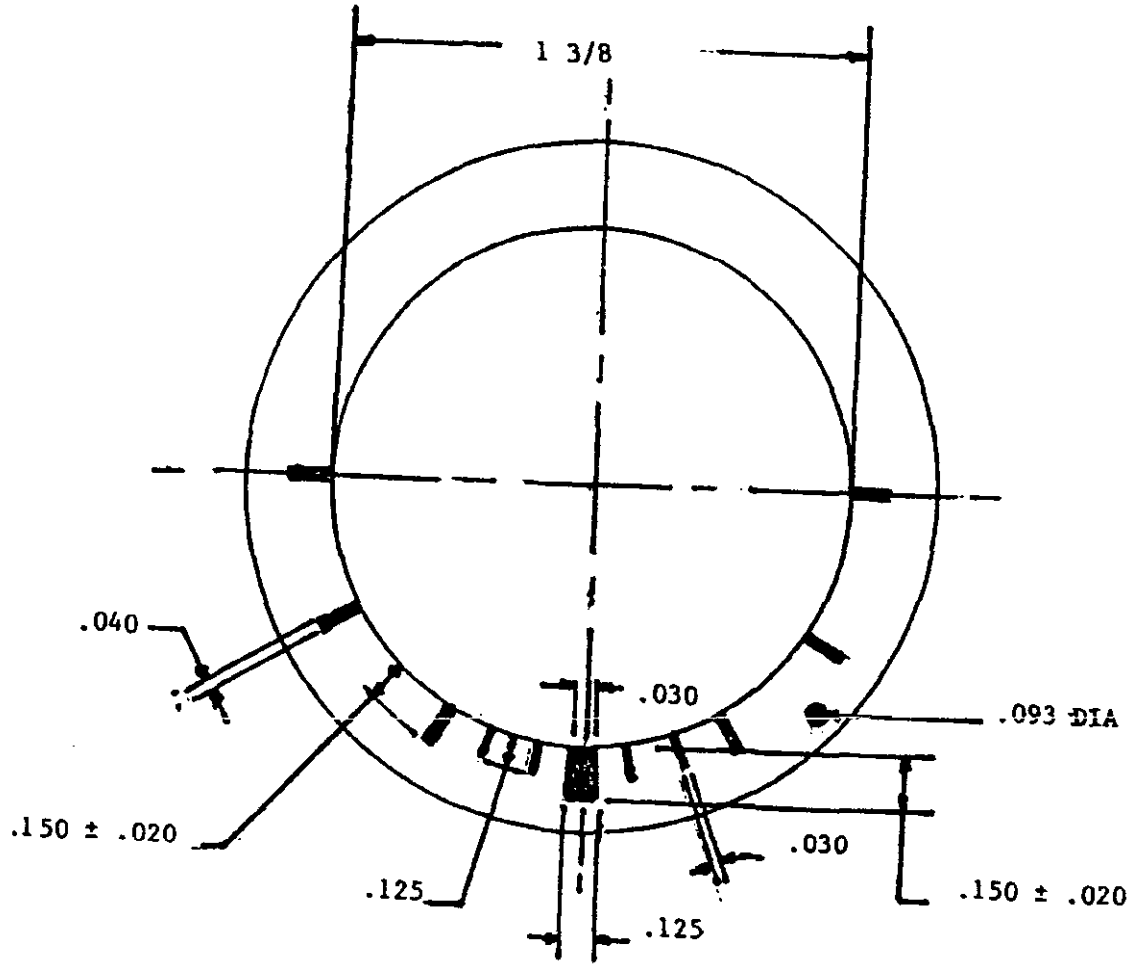
PAGE 8

Figure 2: Delete existing figure 2 and substitute the new figure 2 shown as page 2 of this amendment.

PAGE 11

Add new paragraph 3.11.2.1.

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MS27150-1 OR MS27150-3
(RED & WHITE LIGHTED)
OR MS27150-6 FOR AUGMENTED WHITE LIGHT

DIMENSIONS IN INCHES UNLESS OTHERWISE
SPECIFIED,
TOLERANCES: DECIMALS ± .005
FRACTIONS ± 1/64

FIGURE 2. Bank angle scale details lower location.

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"3.11.2.1 Lighting pins for MS27150-6 assembly. Equip all lamp sockets with white lamps. Derive power for lamps from pin K and pin T of J1 which shall be connected with a jumper wire. Common connection will be pin L."

PAGE 21

3.34.2 Delete and substitute the following:

"3.34.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

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4.6.19 Delete and substitute the following:

"4.6.19 Vibration error. The indicator shall be connected to the test transmitter and subjected to a vibration error test in accordance with MIL-STD-810, Method 514.2, equipment category b.2, table 514.2-IIA, procedure IA, figure 514.2-2A, curve A except that Wo shall be $0.02G^2/Hz$ and the low frequency power spectral density shall be factored by 4db/octave to $0.01G^2/Hz$. The indicator shall meet the following test which shall be conducted at three indications."

PAGE 33

4.6.28, First sentence: Delete "method 501" and substitute "methods 501.1".

4.6.29 Delete the first sentence and substitute the following: "The indicator shall be subjected to a temperature-Altitude test in accordance with MIL-STD-810, method 514.1, equipment category 6, except that examination shall be performed at the end of 72 hours."

4.6.30, Line 2: Delete "method 503" and substitute "method 503.1".

4.6.33, Line 1 and 2: Delete "method 507" and substitute "method 507.1".

4.6.34, Line 2: Delete "method 508" and substitute "method 508.1".

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4.6.36, Line 2: Delete "method 509" and substitute "method 509.1".

4.6.37, Line 2: Delete "method 510" and substitute "method 510.1".

4.6.38, Line 2: Delete "method 516" and "figure 516.1-1" and substitute "method 516.2" and "figure 516.2-1".

4.6.39 Delete and substitute the following"

"4.6.39 Vibration failure. With no power applied, the indicator shall be subjected to a vibration failure test in accordance with MIL-STD-810, method 514.2,

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equipment category b.2, table 514.2-IIA, procedure IA, figure 514.2-2A, curve A except that Wo shall be $0.04G^2/Hz$ and the low frequency power spectral density shall be factored by 4db/octave to $0.02G^2/Hz$. At the conclusion of this test, the indicator shall be subjected to and shall meet the test specified in 4.6.4 and 4.6.7. There shall be no damage to any part of the indicator as a result of this test."

4.6.42 Delete the basic paragraph and substitute the following:

"4.6.42 Qualification reliability. The six indicators specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781. The table I test conditions matrix for fighter aircraft applies. Vibration stress amplitude shall be in accordance with table B-V for cockpit equipment. (Use q dynamic pressure = 1200 psf and $K = .34 \times 10^{-8}$ for computing Wo.) The acceleration power spectral density profile for fuselage shown in figure B-4 applies. The thermal stress operating temperature range shown in table B-II applies. The temperature and time profile should be intergrated into a smooth curve as shown in figure B-3. Table B-VII shows the cold day ambient temperatures that apply to specific altitudes. Table B-VIII shows hot day ambient temperatures that apply to specific altitudes. Moisture stress as specified in paragraph 50.4.3.4.2 for chamber air humidity applies. There is no requirement to induce moisture into the indicator case. Test plan IIC shall be used to demonstrate a minimum acceptable MTBF of 750 hours. During each operating period, with each indicator properly connected, the following conditions shall exist:"

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4.6.43 Delete and substitute the following:

"4.6.43 Longevity. Testing of three of the six test samples specified in 4.6.42 shall be continued through a longevity test for 2,500 hours on each indicator. The same test conditions shall apply. Testing time accrued on the three samples during reliability testing shall apply to the 2,500 hours. No accept/reject criteria for this test is specified. The evaluation shall be limited to the analysis of all failures indentification of design and pattern failures, and the submission of recommended design changes to the contracting activity."

4.7 Delete and substitute the following:

"4.7 Inspection and Test. Test of Methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

PAGE 36

Delete present Section 5 in its entirety and substitute the following new section:

"5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A of C as specified.

5.1.1 Level A.

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5.1.1.1 Cleaning. Unit shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

Delete paragraph 5.1.1.4 and add:

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator, attitude, remote standby, 2 inch, shall be packaged in quantity unit packs of one each in accordance with Method 1C1 of MIL-P-116. Each unit shall be overboxed in PPP-B-636 carton. Carton shall be large enough to allow for application of sufficient cushioning material, MIL-P-26514, between bag and container, of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop test of MIL-P-116.

5.1.2 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

Delete paragraphs 5.2.1 and 5.2.2 and add:

5.2.1 Level A Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, style A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, and of minimum tare and cube, consistent with the protection required.

5.2.2 Level B. Units packaged as specified in 5.1.1 shall be packed in weather resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows: Indicator, Attitude Remote Standby, 2-inch."

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6.4, last sentence: Delete and substitute the following:

"The activity responsible for the Qualified Products List is OC-ALC/MMEDO

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Tinker AFB OK 73145-5990, and information pertaining to qualification of products may be obtained from that activity."

PAGE 38

Delete paragraph 6.7

The margins of this amendment are marked with a vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Certain provisions of this document are subject to international standardization agreement ASCC AIR STD 10/42C. When amendment, revision, or cancellation of this document is proposed which will modify the international agreement concerned, the preparing activity takes appropriate action through international standardization channels including the Departmental Standardization Offices (DepSo) to change the agreement or make other appropriate accomodations.

Custodian:
Air Force - 99
Army - AV
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Preparing Activity:
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